NVDLA on PetaLinux 2024.1

Correctness, latency, throughput, and monitored energy at a glance

100 / 100

LeNet stability

One boot; no module reload or PL reset

246 / 246

ResNet-50 hardware layers

Exact match to source-built nv_small VP golden

1.34x

CPU INT8 execution advantage

378.7 ms CPU INT8 vs 507.7 ms NVDLA on ResNet-50

1.92x

NVDLA cold-deployment advantage

1.336 s NVDLA vs 2.566 s CPU INT8 on ResNet-50

-62.3%

NVDLA incremental energy

0.134 J NVDLA vs 0.356 J CPU INT8 on ResNet-50

20 x 2

Input-sensitivity control

Balanced image sets; three fresh boots per model

Comparison boundary: nv_small NVDLA INT8 is compared with both FP32 and independently quantized QDQ INT8 ONNX Runtime on four Cortex-A53 cores. Equal nominal precision does not imply identical quantization or graph transformation. The primary campaign contains 60 sessions; six supplementary sessions test 20 images per model.

Correctness before performance

Layered acceptance

- OK Pinned inputs
- OK ABI and build
- OK Verified nv_small VP
- OK Driver probe
- OK GEM mapping
- OK IRQ + engine completion
- OK Exact tensor + repeat

LeNet / MNIST

Fast repeat oracle

INPUT **1 x 1 x 28 x 28**	LOADABLE **0.446 MB**	HARDWARE LAYERS **10**
ENGINE MIX **4 Conv / 4 SDP / 2 PDP**	VP **Exact output**	BOARD **Exact; 100/100 repeats**

Output: 0 2 0 0 0 0 0 124 0 0

ResNet-50

Large multi-engine graph

INPUT **1 x 3 x 224 x 224**	LOADABLE **25.77 MB**	HARDWARE LAYERS **246**
ENGINE MIX **114 Conv / 130 SDP / 2 PDP**	VP **246/246; golden established**	BOARD **246/246; exact hash**

Output: 1,000 signed values; SHA-256 842d34f...

Sources: VP configuration audit, differential trace, board stability and ResNet-50 artifacts

Correctness-qualified samples; no outliers discarded

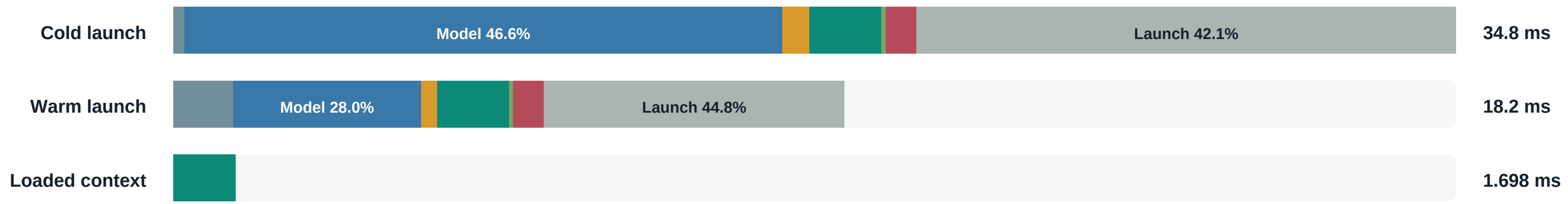
Where does NVDLA deployment time go?

Initialization | Model loading | Input + buffers | Runtime execution | Result handling | Teardown | Launch / unprofiled

LeNet

bar length is relative to this model's cold launch-to-exit time

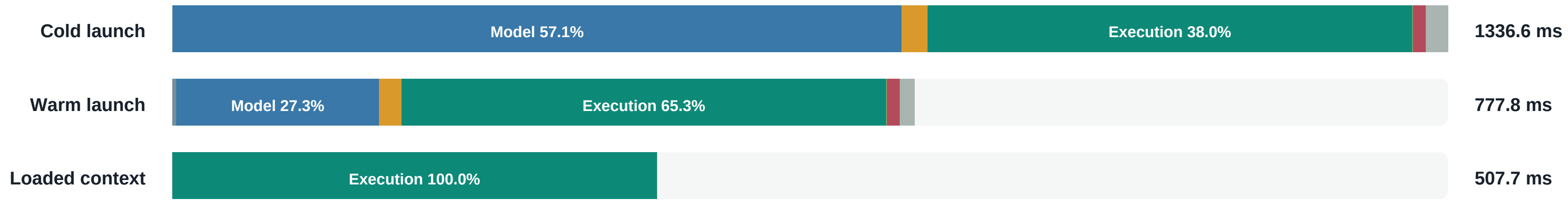
Warm runtime execution share 10.7% | loaded context 1.698 ms



ResNet-50

bar length is relative to this model's cold launch-to-exit time

Warm runtime execution share 65.3% | loaded context 507.7 ms

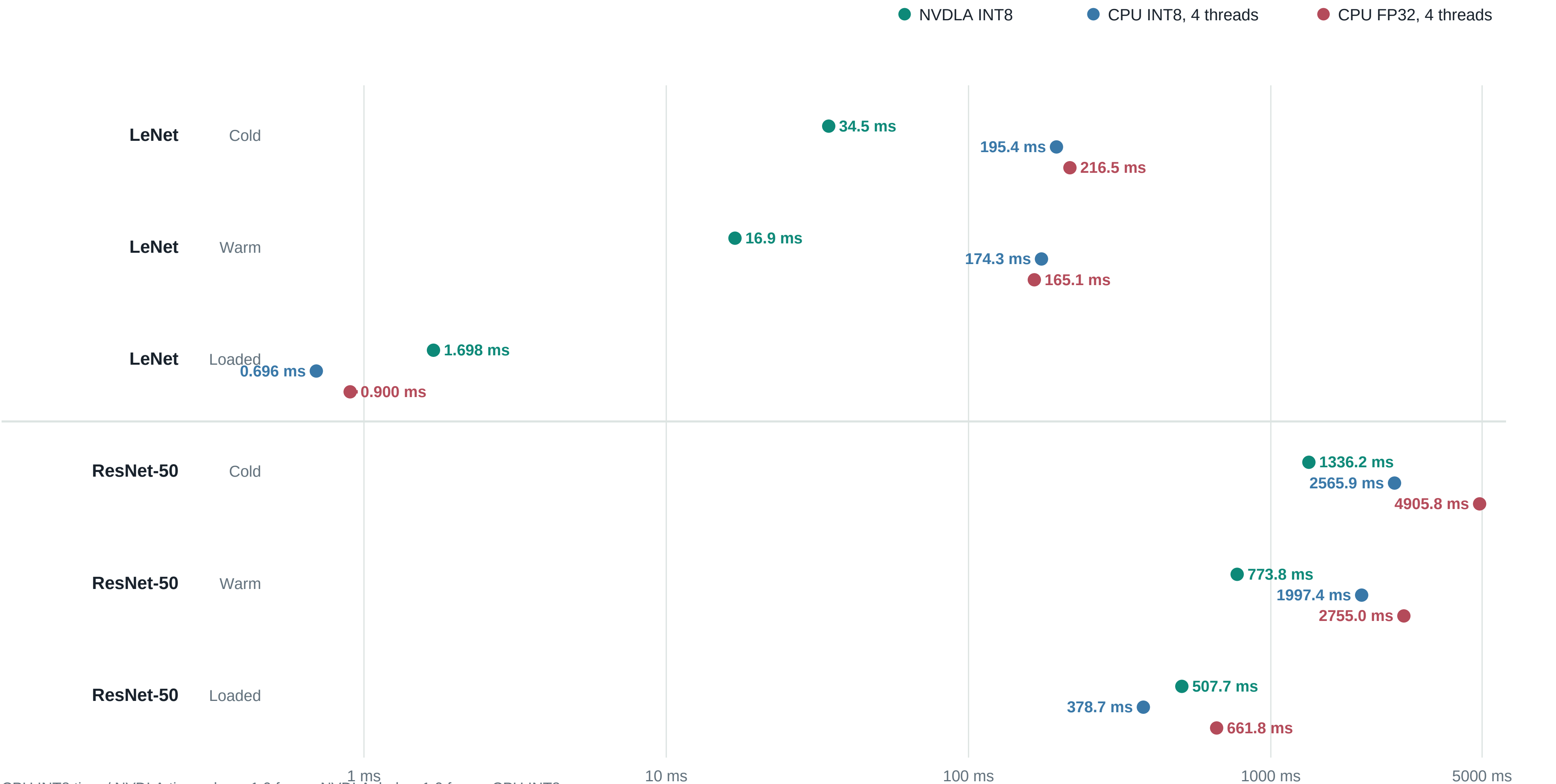


Timing boundary: cold and warm bars span process launch to exit. Loaded context reuses the model and buffers; its blocking runtime execution includes userspace dispatch, ioctl/KMD scheduling, accelerator work, IRQ completion, and emulator tasks.

Phase means from five fresh-boot latency sessions; no power sampling

Correctness-qualified samples; no outliers discarded

Three deployed stacks across each timing boundary



CPU INT8 time / NVDLA time: above 1.0 favors NVDLA; below 1.0 favors CPU INT8

5.67x
LeNet cold

10.33x
LeNet warm

0.41x
LeNet loaded

1.92x
ResNet cold

2.58x
ResNet warm

0.75x
ResNet loaded

Unpowered cohorts; points are medians across five independent boot-session medians

Correctness-qualified samples; no outliers discarded

Does execution time depend on the image?

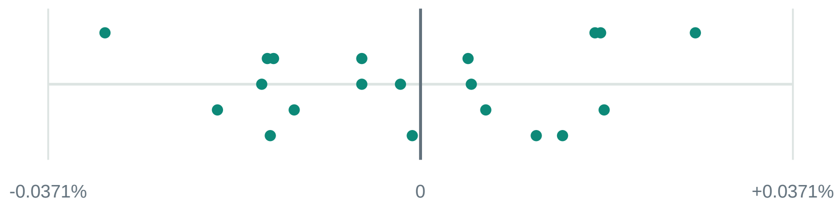
Execution-validity result: every input produced a repeat-stable output, increased IRQ activity, and completed without a bad kernel pattern.

LeNet / MNIST

20 inputs x 30 measured executions; loaded model and buffers reused

Per-input median execution deviation

model-specific horizontal scale



Observed range across the 20 medians: 0.0589%

Input decode and file I/O are outside this execution interval.

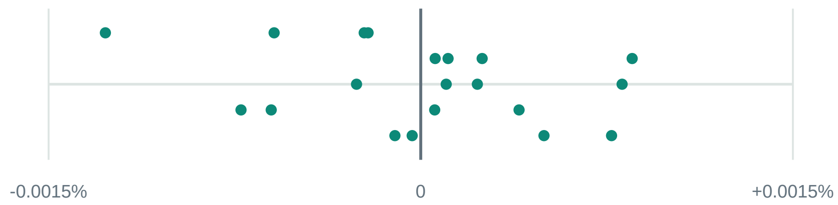
1.699 ms	0.002 ms	20 / 20	16 / 20
Median execution	Prepared input copy	Stable outputs	Recorded top-1

ResNet-50 / Imagenette

20 inputs x 15 measured executions; loaded model and buffers reused

Per-input median execution deviation

model-specific horizontal scale



Observed range across the 20 medians: 0.0022%

Input decode and file I/O are outside this execution interval.

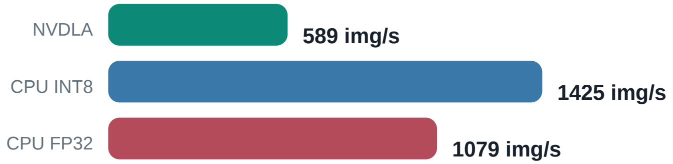
507.8 ms	0.171 ms	20 / 20	17 / 20
Median execution	Prepared input copy	Stable outputs	Recorded top-5

Interpretation: class accuracy is descriptive model-quality context from a small curated set. Hardware acceptance is based on repeat-stable output, IRQ progress, and clean kernel logs.

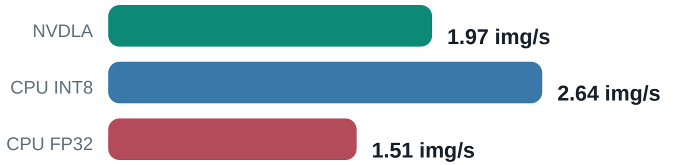
What changes between LeNet and ResNet-50?

Loaded-context throughput

LeNet



ResNet-50



Workload scale and operation mix

LeNet

NVDLA: 0.446 MB loadable | 10 hardware layers

CPU ONNX: 1.73 MB FP32 / 0.45 MB INT8

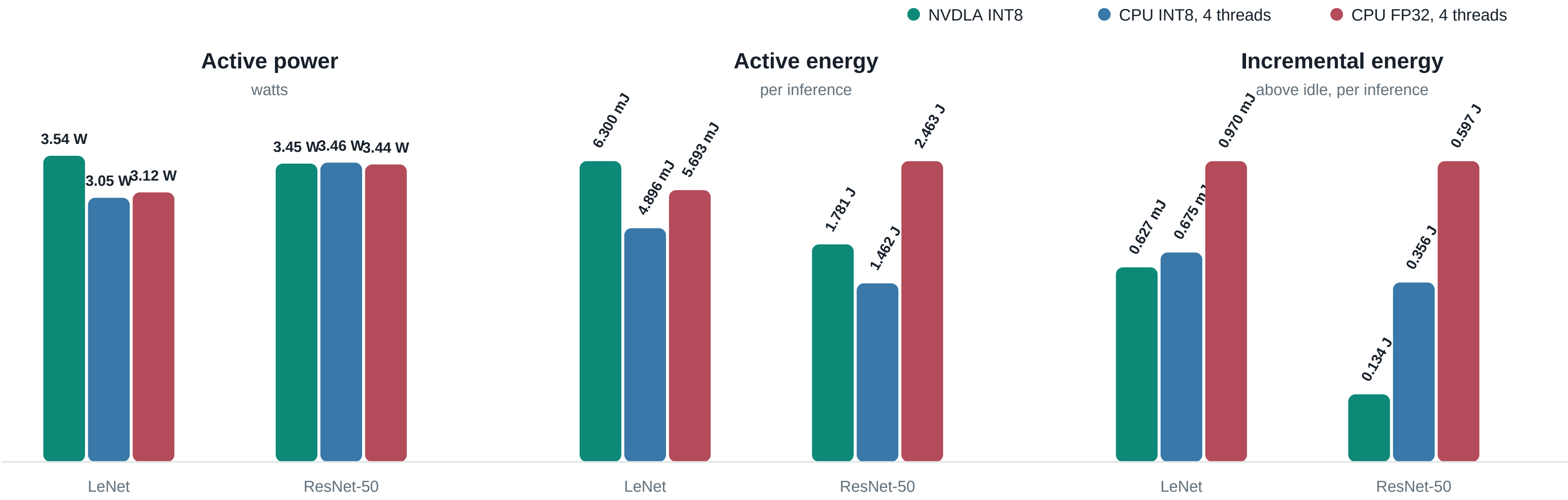
ResNet-50

NVDLA: 25.77 MB loadable | 246 hardware layers

CPU ONNX: 102.48 MB FP32 / 26.09 MB INT8

ResNet-50 has **24.6x** more hardware layers and a **57.8x** larger loadable.
Its larger execution graph amortizes fixed submission and interrupt costs.

Monitored PS + PL rails during inference



LeNet

+28.7% NVDLA active energy

-7.1% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

ResNet-50

+21.8% NVDLA active energy

-62.3% incremental

Change is NVDLA relative to CPU INT8. Active includes the monitored platform; incremental subtracts driver-loaded idle.

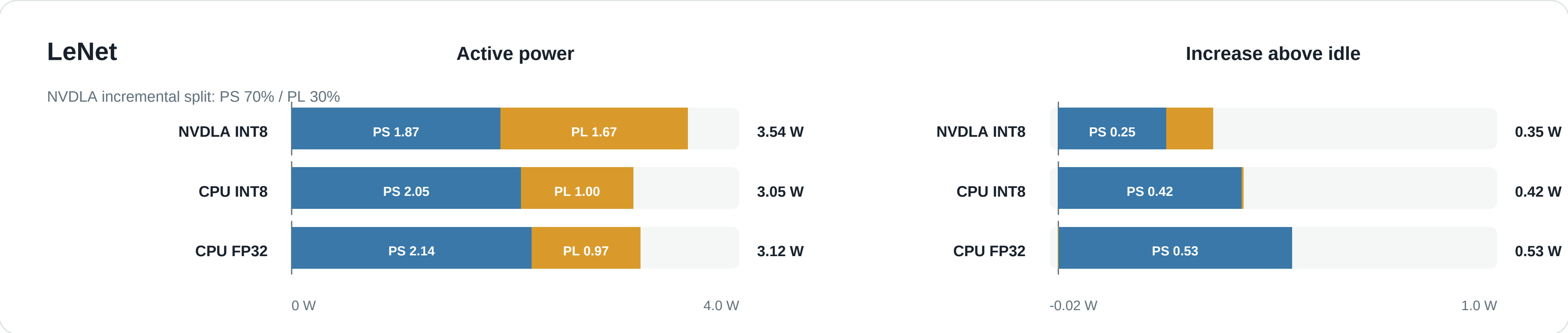
Power cohorts are separate from primary latency cohorts; 50 ms sampling with endpoint capture

Correctness-qualified samples; no outliers discarded

How do PS and PL contribute?

PS rails   PL rails

Active = complete monitored board state | Incremental = active minus stack-specific idle



Reading the split: CPU inference activity is overwhelmingly on PS rails. NVDLA distributes activity across PS and PL; for ResNet-50, PL contributes about 62% of its incremental power. Tiny negative PL deltas are retained and indicate the sensor/idle-subtraction noise floor.

Five fresh-boot powered sessions per model and stack; means across monitored rails

Correctness-qualified samples; no outliers discarded

What was collected, and how to read it

1. **Fresh boot** — Unique Linux boot ID
2. **Settle + verify** — Clock, frequency, hashes
3. **Correctness gate** — Golden output + kernel health
4. **Measure** — Unpowered latency or powered batch
5. **Archive** — Raw profiles, rails, logs
6. **Aggregate** — Five session medians + 95% CI

Metric inventory

Correctness	Output hashes, exact tensors, engine sequence, IRQ deltas, repeat failures
Latency	Cold deployment, warm deployment, loaded-context execution, runtime phases
Power	18 PS + PL rails, idle/active/incremental watts, integrated energy/inference
Statistics	Raw samples, mean/median, spread, percentiles, retained outliers, bootstrap CI
Workload	Input shape, loadable/model size, HWL count, per-engine operation count
Throughput	Images/s for every timing boundary; CPU/NVDLA latency ratios
Environment	Kernel, binary hashes, clock, CPU affinity/frequency/governor, boot ID
Evidence	Serial, dmesg, runtime profiles, sensor samples, manifests, selection hashes

Interpret with: one ZCU102 and nv_small implementation; NVDLA INT8 versus independently quantized CPU INT8 plus an FP32 reference; monitored rails rather than wall input; five boots per primary cohort and three per input-sensitivity model.

Full provenance and raw samples remain in artifacts/final-reports and the selected session archives

Correctness-qualified samples; no outliers discarded